CHAPTER

33

The z-Transform

Just as analog filters are designed using the Laplace transform, recursive digital filters are developed with a parallel technique called the z-transform. The overall strategy of these two transforms is the same: probe the impulse response with sinusoids and exponentials to find the system's poles and zeros. The Laplace transform deals with differential equations, the s-domain, and the s-plane. Correspondingly, the z-transform deals with difference equations, the z-domain, and the z-plane. However, the two techniques are not a mirror image of each other; the s-plane is arranged in a rectangular coordinate system, while the z-plane uses a polar format. Recursive digital filters are often designed by starting with one of the classic analog filters, such as the Butterworth, Chebyshev, or elliptic. A series of mathematical conversions are then used to obtain the desired digital filter. The z-transform provides the framework for this mathematics. The Chebyshev filter design program presented in Chapter 20 uses this approach, and is discussed in detail in this chapter.

The Nature of the z-Domain

To reinforce that the Laplace and z-transforms are parallel techniques, we will start with the Laplace transform and show how it can be changed into the z-transform. From the last chapter, the Laplace transform is defined by the relationship between the time domain and s-domain signals:

$$X(s) = \int_{t=-\infty}^{\infty} x(t)\, e^{-st}\, dt$$

where $x(t)$ and $X(s)$ are the time domain and s-domain representation of the signal, respectively. As discussed in the last chapter, this equation analyzes the time domain signal in terms of sine and cosine waves that have an exponentially changing amplitude. This can be understood by replacing the